

03
CHAPTER

39V432

Introduction to Database Management System

LEARNING OBJECTIVES

To understand database concepts, components and its functions.

- **To know about relational model of data**
- **To understand Query languages for databases.**
- **Enables to write SQL commands and query processing**
- **To enhance the programming skills and Techniques using MySQL**

3.1 Introduction to Database Management System

DBMS stands for Database Management System, so let us break down the words what they really mean. A database is a place where we store, retrieve and manage data. So what's a data then? Meaningful information like your name, your favorite color etc to complex data like astronomical data that scientist handle, everything comes under database. The management system refers to a set of programs to manage the data, we have with various actions like storing, retrieving, filtering etc. Some of the popular Database Management System is MySQL, Oracle etc. Giving protection to data, user-friendly for users etc, are some of the notable features of good DBMS.

3.1.1 Introduction to DBMS

Definition: "A database management system (DBMS) is system software for creating and managing databases. The DBMS provides users and programmers with a systematic way to create, retrieve, update and manage data."

What type of data is stored in a database?

In a database, we would be grouping only related data together and storing them under one group name called table. This helps in identifying which data is stored where and under what name.

3.1.2 Evolution of DBMS

The concept of storing the data started before 40 years in various formats. In earlier days they have used punched card technology to store the data. Then files were used. The file systems were known as predecessor of database system. Various access methods in file system were indexed, random and sequential access. The file system had more limitations like

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